# MIAE 215 Quiz #1

Please write the following programs by hand using a pen / pencil and paper. Then scan in your solutions (using a scanner) or take pictures of them (using a cell phone / camera) and submit them on Moodle.

You have from 8:45 am to 9:50 am to complete the quiz and another 10 minutes to scan and submit the quiz. All submissions are due at 10 am. Late quizzes will not be accepted.

If you are not sure about a question, please state your assumptions / interpretation and proceed to solve the problem based on your assumptions.

1) Write a program that inputs 3 real numbers from the keyboard. It then calculates and prints out the average of the 3 numbers. If an input is greater than 7.5 the program prints out “input is greater than 7.5”.

2) Write a program that calculates a table of function values with 11 rows and 3 columns. Your program should exactly produce the following output:

| x | sin(x) | exp(x) |
|---|---|---|
| 0.00000e0 | 0.00000e0 | 1.00000e0 |
| 1.00000e-1 | 9.98334e-2 | 1.10517e0 |
| 2.00000e-1 | 1.98669e-1 | 1.22140e0 |
| … | ... | … |
| … | … | … |
| 1.00000e0 | 8.41471e-1 | 2.71828e0 |

Hint the control character for tab is ‘\t’.

3) Write a program that determines the largest value possible for an unsigned long integer by incrementing an unsigned long integer i by one until the largest value is achieved. Then print out the largest value to the screen. Hint: use a double x in your for loop since it can hold very large numbers. What is the smallest value possible for i ? What is the size in bytes of i ?